

01 Three resistors A, B and C are connected in a circuit with a cell, as shown in Fig. 1.1.

Cell is in series with A
& Parallel with B & C

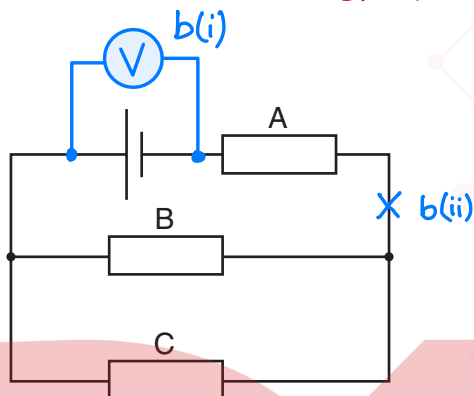


Fig. 1.1

(a) State the arrangement of resistors B and C in the circuit.

Parallel

[1]

(b) On Fig. 1.1, draw

(i) the symbol for a voltmeter to measure the voltage provided by the cell,

[1]

(ii) an X to indicate the position of an ammeter to measure the current in resistor A.

[1]

(c) The voltmeter is shown in Fig. 1.2.

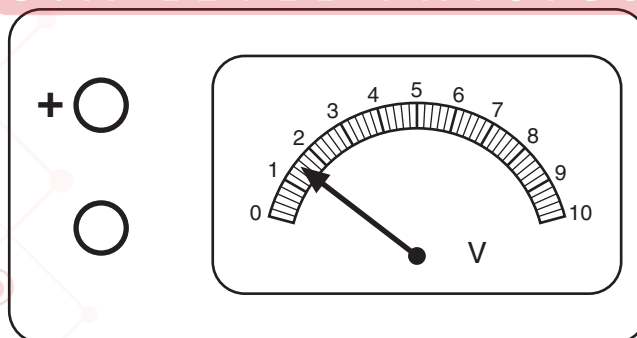


Fig. 1.2

(i) State the voltmeter reading V on Fig. 1.2.

V = 1.5 V [1]



- (ii) Resistors A, B and C have resistances R_A , R_B and R_C where $R_A = R_B = R_C = 10\Omega$.

Use your answer for V in (c)(i) and the relationship below to find the value of the current I in resistor A.

$$\frac{V}{I} = R_A + \frac{R_C R_B}{R_C + R_B}$$

$$\frac{1.5}{I} = 10 + \frac{10 \times 10}{10 + 10}$$

$$\frac{1.5}{I} = 10 + 5$$

$$\frac{1.5}{I} = 15$$

$$I = \frac{1.5}{15}$$

$$I = 0.1 \text{ A} \dots\dots\dots [1]$$

- (d) Four students try to build the circuit of Fig. 3.1 with three 10Ω resistors and a cell. Their circuit diagrams are shown in Fig. 1.3.

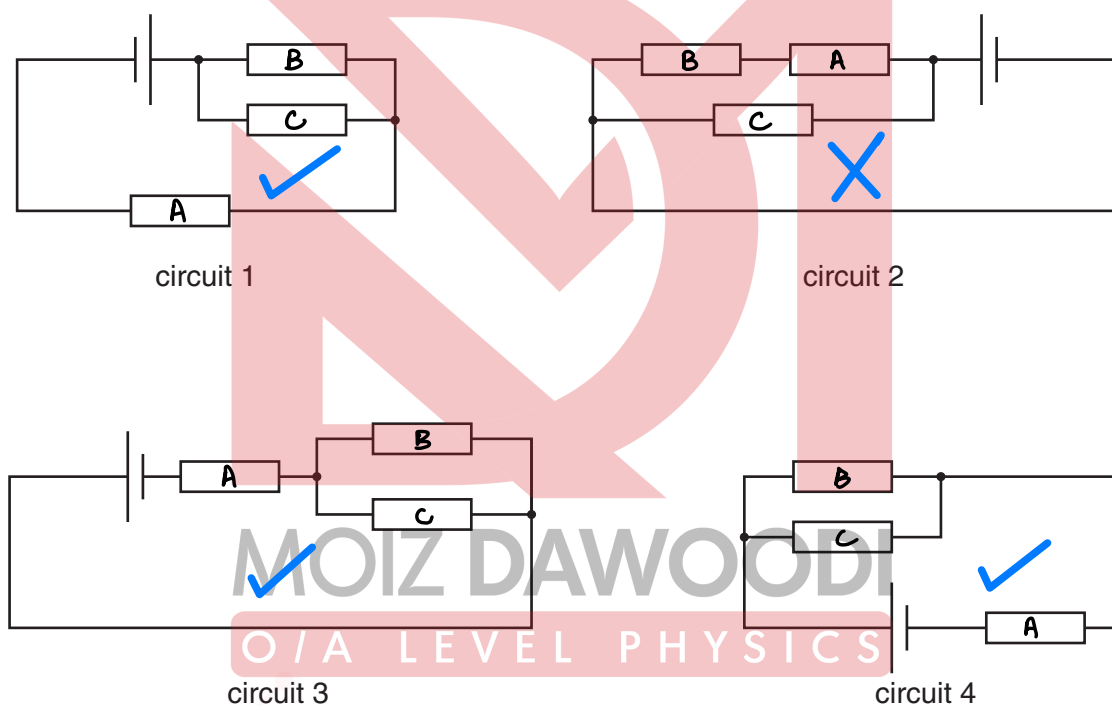


Fig. 1.3

State and explain which circuit is **not** the same as the circuit in Fig. 1.1.

Circuit 2 is not same as two resistors are in series while two resistor needs to be in parallel. [1]

[MJ2011/P42/Q3]

- 02 A cathode-ray oscilloscope (c.r.o.) is used to check the voltage of a cell. Fig. 2.1 shows the cell in a simple circuit and also the front of the c.r.o.

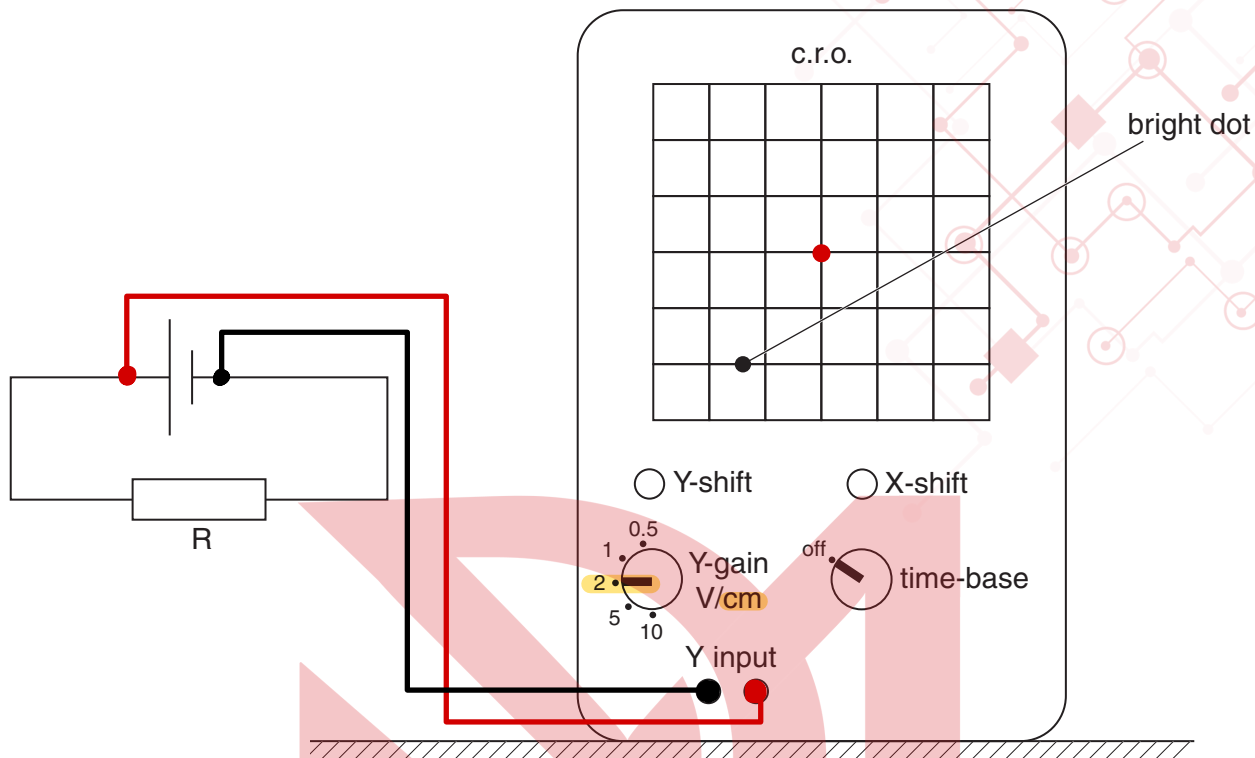


Fig. 2.1

The c.r.o. is switched on and the time-base is switched off. A bright dot is seen on the screen, as shown in Fig. 2.1.

- (a) State which controls are used to move the dot to the centre of the screen.

X-shift & Y-shift

.....[1]

- (b) On Fig. 2.1, draw connections from the circuit to the c.r.o. so that the voltage of the cell can be measured. [1]

- (c) The dot is now in the centre of the screen.
Fig. 2.2 shows the dot on the screen of the c.r.o. before and after the cell is connected.

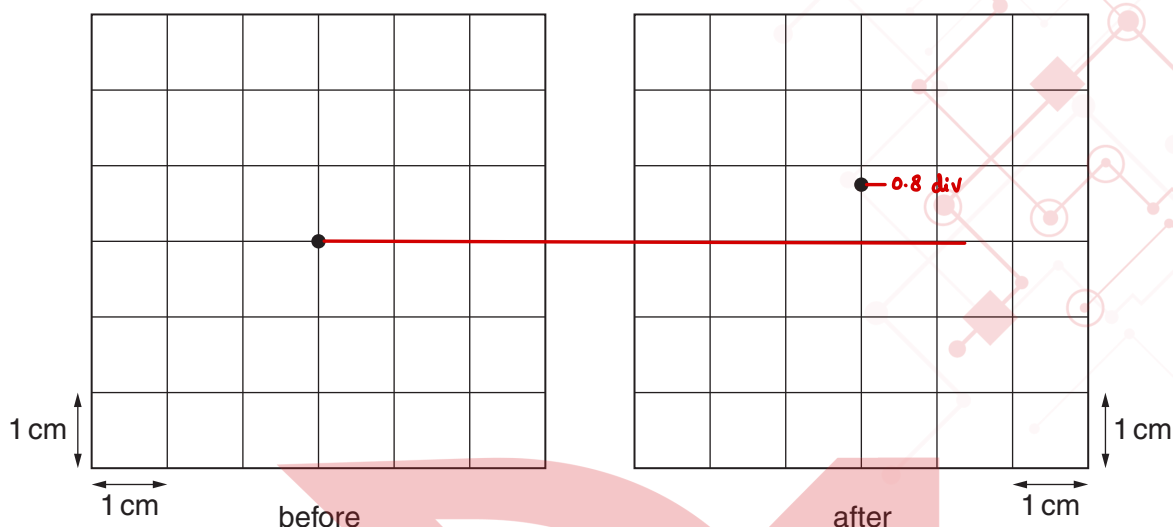


Fig. 2.2

- (i) Use Fig. 2.1 to state what is represented by 1 cm on the vertical axis on the c.r.o.
..... 2.0 V [1]
- (ii) Determine the voltage of the cell.
 $1\text{ cm} = 2\text{ V}$
 $0.8\text{ cm} = V$
 $V = 0.8 \times 2$
 voltage = 1.6 V [1]
- (d) Describe and explain what is observed when
- (i) the connections from the circuit to the c.r.o. are reversed,
 Trace moves downward at -1.6 V .
 [1]
- (ii) the time-base is switched on.
 Horizontal trace
 [1]

[ON2011/P41/Q3]

03 A teacher demonstrates the force on a wire carrying a current in a magnetic field.

Fig. 3.1 shows some of the apparatus used.

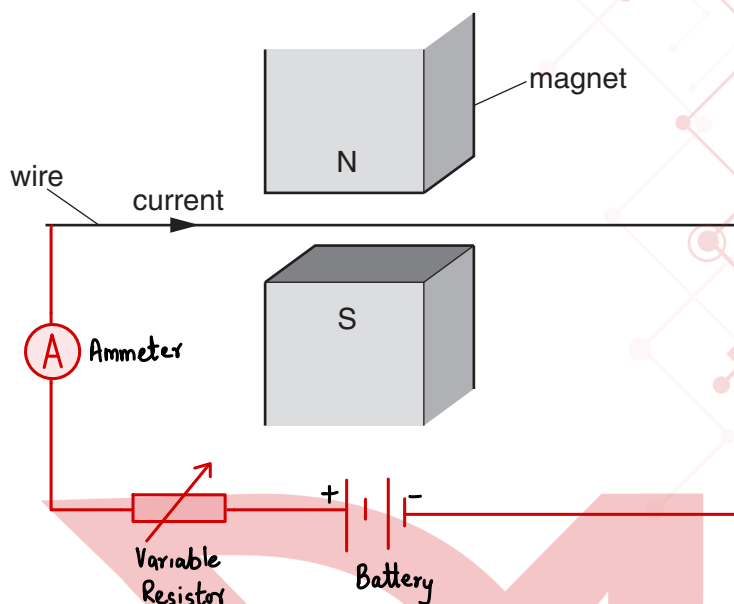


Fig. 3.1

(a) The current is to be varied and measured.

(i) On Fig. 3.1, draw the circuit diagram of the circuit used to provide the current. Include a method to vary and measure the current. [2]

(ii) Describe how to increase the current in the wire.

Decrease the resistance of variable resistor.

[1]

(iii) Describe how to reverse the direction of the current in the wire.

Reverse the connections from battery.

[1]

(iv) Describe how to reverse the direction of the magnetic field.

Reverse the poles of magnet.

[1]

(b) The current is increased to a very large value. Suggest one problem this may cause.

Circuit may get overheated & eventually will catch fire.

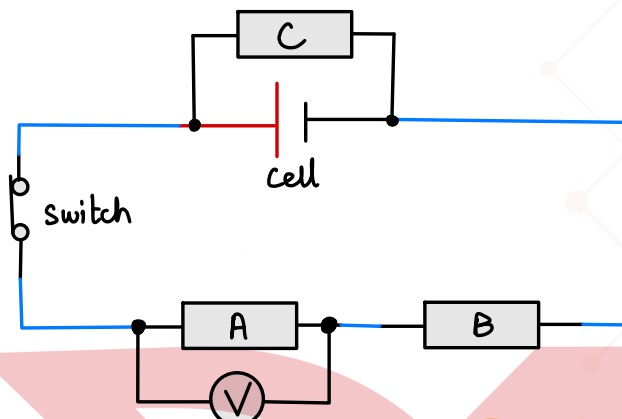
[1]

[ON2011/P42/Q2]

- 04 (a) A student connects a cell, a switch and three resistors to make a circuit. The resistors are labelled A, B and C.

Resistors A and B, the cell and the switch are all in series.
Resistor C is in parallel with the cell.

- (i) In the space below, draw the circuit diagram.



[2]

- (ii) On your circuit diagram, draw the symbol for a voltmeter, connected to measure the voltage across resistor A. [1]

- (b) A student connects the circuit shown in Fig. 4.1. X, Y and Z are three identical lamps.

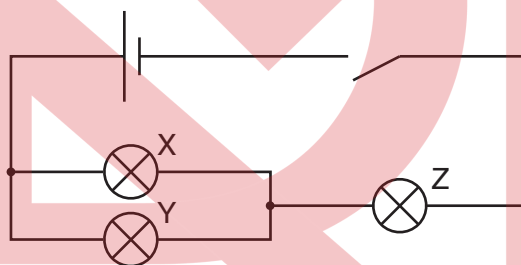


Fig. 4.1

The student closes the switch. Lamp Z lights dimly. Lamp X and lamp Y do not light.

Tick TWO of the following which are possible explanations.

both lamp X and lamp Y are faulty	
only lamp X is faulty	
only lamp Y is faulty	
the cell is running down	✓
a connecting lead from the cell is broken	
the current in lamp X and in lamp Y is too small	✓

[2]

[MJ2012/P42/Q3]

- 05 (a) A student finds an old magnet at the back of a drawer containing other magnets. He designs an experiment to find out if it is still magnetised.
- (i) He brings a plotting compass near to end A of the old magnet, as shown in Fig. 5.1.

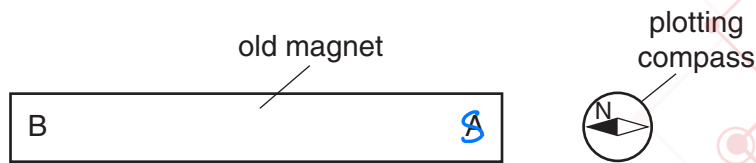


Fig. 5.1

State the polarity of end A of the old magnet.

South

[1]

- (ii) The plotting compass is then brought near to end B, as shown in Fig. 5.2.



Fig. 5.2

State the polarity of end B.

South

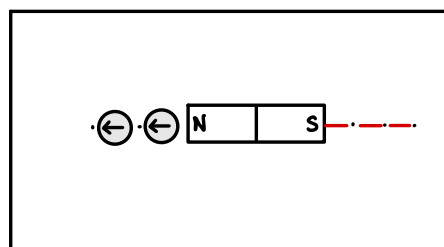
[1]

- (iii) Suggest a possible explanation for the student's results.

Magnet has lost its magnetism

[1]

- (b) Describe how the student can use the plotting compass to plot the shape of the magnetic field around a new magnet. You may use a diagram in your explanation.



Place compass on paper & draw an outline around it.

Place compass near north of magnet & mark a dot where compass needle alligns. Replace compass such that it alligns with dot & remark another. Join all the dots.

[3]

[MJ2012/P42/Q4]

- 06 A student investigates the effect of changing the length of a piece of resistance wire in a circuit containing a lamp. The resistance of the lamp is similar to the resistance of a 1 m length of the wire.

The circuit is set up as shown in Fig. 6.1.

$$R_{\text{lamp}} = R_{(1\text{m})}$$

$$R \propto L$$

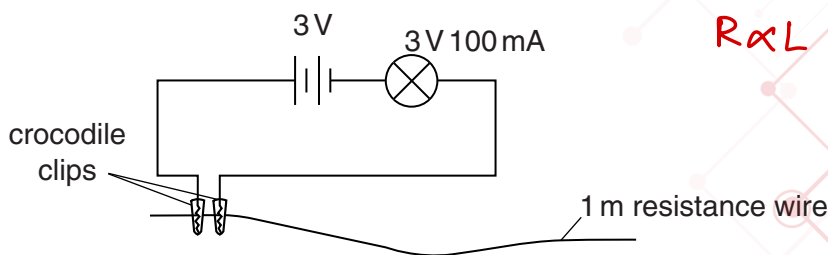


Fig. 6.1

- (a) The student checks that the circuit is working by touching together the two crocodile clips.

- (i) State what the student observes if the circuit is working correctly.

Lamp glows [1]

- (ii) Describe one possible fault that prevents the circuit working correctly.

Cell may be run down or broken wire. [1]

- (b) When the circuit is working correctly, the student connects the two crocodile clips close together on the wire, as shown in Fig. 6.1.

- (i) Describe what the student observes as the crocodile clips are slowly moved further apart on the wire.

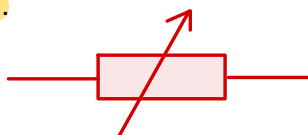
As length increases so resistance also increases hence lamp glows dimly [1]

- (ii) A circuit component is based on changing the length of the wire between two contacts.

1. State the name of this component.

Variable Resistor / Rheostat [1]

2. Draw its circuit symbol.



[1]

3. The component contains a long length of wire. Explain how the component is made small.

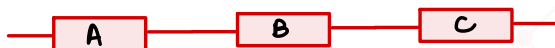
Make it in the shape of coil. [1]

[ON2012/P41/Q2]

07 A student investigates the behaviour of resistors.

(a) Three resistors A, B and C are labelled 100Ω , 150Ω and 220Ω respectively.

- (i) 1. Draw the arrangement of A, B and C that gives the largest possible total resistance.



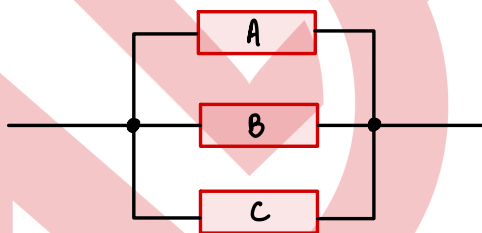
[1]

2. Calculate the resistance of this arrangement.

$$100 + 150 + 220$$

resistance = 470Ω [1]

- (ii) Draw the arrangement of A, B and C that gives the smallest possible total resistance.



[1]

[MJ2013/P41/Q2]

08 A solar cell converts light energy into electrical energy.

A student investigates the maximum e.m.f. produced by a solar cell in the laboratory.

Fig. 8.1 shows the symbol for a solar cell.

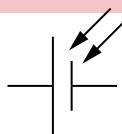
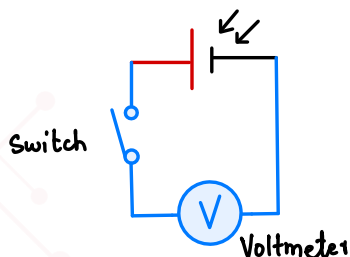


Fig. 8.1

(a) The student uses a voltmeter to measure the e.m.f. produced by the solar cell.

- (i) Draw a diagram of the circuit he uses. Include a switch in the circuit.



[1]

- (ii) When the student closes the switch, he notices that the voltmeter needle moves backwards, as shown in Fig. 8.2.

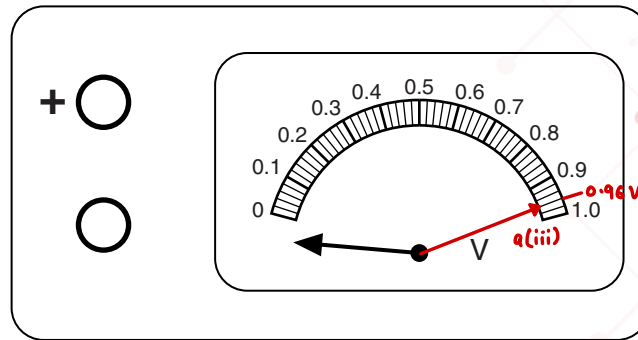


Fig. 8.2

Explain why this happens and how the student can correct this.

This is due to zero error in voltmeter. Student measures this zero error & subtracts this from the final reading. [2]

- (iii) The student corrects the problem and, when the switch is closed, the voltmeter now reads 0.96 V. On Fig. 8.2, mark the new position of the needle. [1]

- (b) To investigate the solar cell, the student uses light entering the laboratory through a window.

When the student moves his head to read the voltmeter, there is a large decrease in the reading.

Suggest a reason for this, and explain how the student can prevent this happening when he moves.

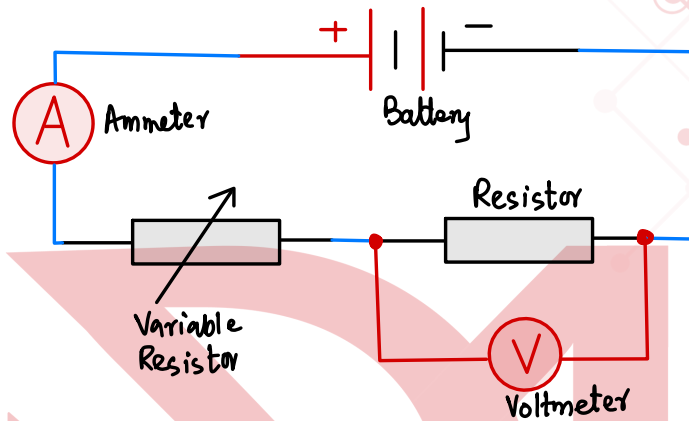
Shadow of his head is on the solar cell. Stand at the opposite side of the window. [2]

[MJ2013/P42/Q4]

- 09 (a) Describe an experiment to measure the resistance of a resistor using an ammeter and a voltmeter.

Your account should include

- a circuit diagram,
- the readings taken,
- a method of taking a range of readings,
- a method of determining the resistance from the readings.



Ammeter in series measures current & voltmeter in parallel measures voltage of resistor. Multiple readings are taken by changing resistance of variable resistors & voltage, current readings are recorded. Using formula " $R = V/I$ ", resistance is found.

[4]

- (b) Describe one way to improve the accuracy of the result.

Use digital meters / Avoid parallax error by keeping eyes perpendicular to scale / Take average resistance.

[1]

[ON2013/P41/Q4]

10 A student uses a lemon to make a simple electric cell.

He inserts a copper strip and an iron nail into the lemon, as shown in Fig. 10.1.

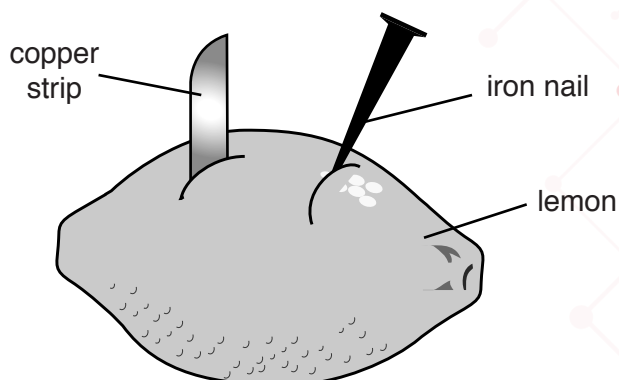


Fig. 10.1

The student connects a centre-zero voltmeter between the copper strip and the iron nail to measure the e.m.f. produced by the lemon cell.

(a) Fig. 10.2 shows the voltmeter with the reading obtained.

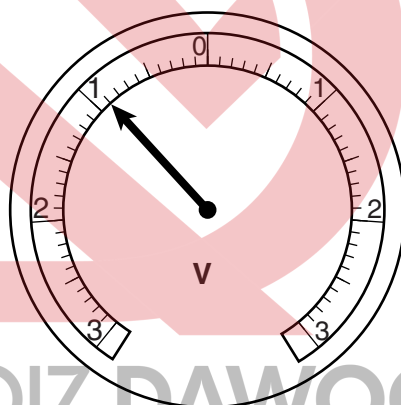


Fig. 10.2

(i) Record the e.m.f. shown on the voltmeter.

e.m.f. = -0.9V [1]

(ii) Suggest how the connecting leads from the voltmeter can be securely attached to the strip and to the nail.

Using crocodile clips. [1]

(iii) State what happens if the connections to the voltmeter are reversed.

Needle will now move to right +0.9V. [1]

- (b) Suggest one reason why lemons are **not** used to power simple electronic equipment such as **digital watches**.

Not enough emf.

[1]

- (c) (i) The student connects three lemon cells in **series** and measures the **total e.m.f.** with the voltmeter.

1. State the total e.m.f. that you would expect.

-0.9×3

e.m.f. = -2.7 V [1]

2. On Fig. 10.3, draw the connections that the student uses.

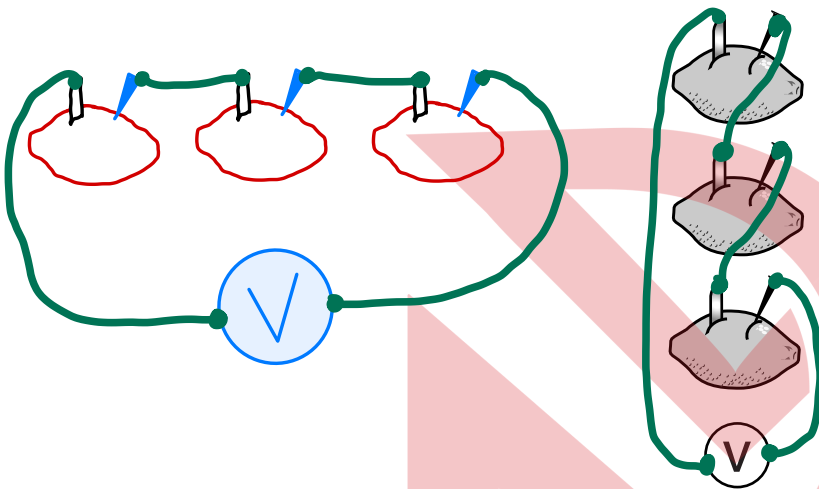


Fig. 10.3

[1]

- (ii) The student connects **three lemon cells** in **parallel** and measures the **total e.m.f.** with the **voltmeter**.

1. State the **total e.m.f.** that you would expect.

e.m.f. = -0.9 V [1]

2. On Fig. 10.4, draw the connections that the student uses.

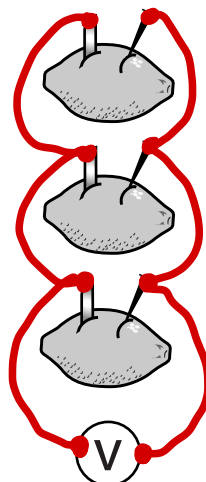
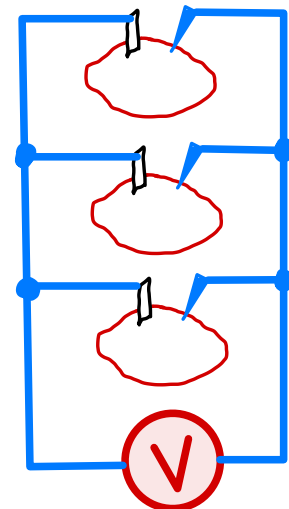


Fig. 10.4



[1]

[ON2013/P42/Q3]

11 A student uses a voltmeter.

- (a) State the quantity measured with a voltmeter.

Voltage/emf/p.d.

[1]

- (b) Fig. 11.1 shows an analogue voltmeter.

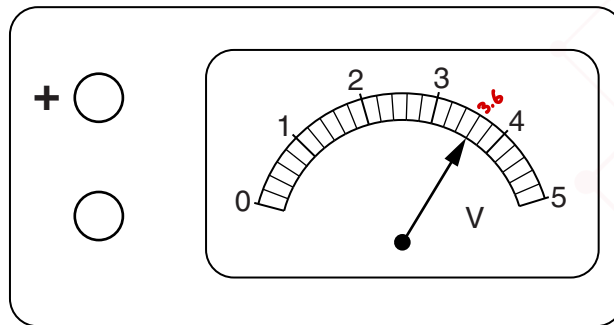


Fig. 11.1

State the reading on the voltmeter.

reading = 3.6 V [1]

- (c) A school has both digital and analogue voltmeters. Suggest one advantage of using a digital voltmeter rather than an analogue voltmeter.

Digital meters are more accurate.

[1]

- (d) The student is asked to connect three 2.0V cells in series with a resistor R of resistance 100Ω. The student sets up the circuit as shown in Fig. 11.2.

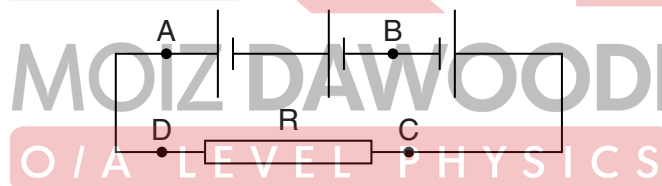


Fig. 11.2

- (i) State the reading on the voltmeter when it is connected across

1. AD, Same points reading = 0 V [1]

2. AB, 2 + 2 reading = 4.0 V [1]

3. DC, 2 + 2 - 2 reading = 2.0 V [1]

- (ii) R is replaced by a resistor of resistance 1000Ω. There is no change in the reading on the voltmeter across DC. Explain this.

(V = IR) As resistance increases by the factor of 10, current decreases by the factor of 10. So P.d remains same [1]

[MJ2014/P41/Q1]

12 A student investigates the use of ammeters in a circuit.

(a) State the quantity measured with an ammeter.

Current [1]

(b) Explain why it is important for an ammeter to have a low resistance.

It is placed in series so that current flows easily. [1]

(c) Fig. 12.1 shows an analogue ammeter.

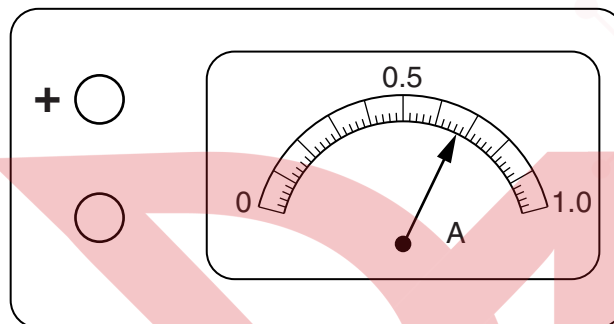


Fig. 12.1

State the reading on the ammeter.

reading = 0.67 A [1]

(d) A school has both digital and analogue ammeters.

Suggest one advantage of using a digital ammeter rather than an analogue ammeter.

More accurate [1]

(e) Fig. 12.2 shows a simple circuit.

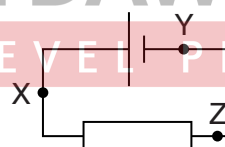


Fig. 12.2

A student connects three similar ammeters at X, at Y and at Z. The ammeters give slightly different readings.

(i) Explain why all the ammeters should give the same reading.

Current remains same in series. [1]

(ii) Suggest a reason for the slight differences in the three readings.

Due to parallax error / zero error. [1]

[MJ2014/P42/Q2]

13 (b) The student uses three identical 24Ω resistors.

- (i) Describe in words how the three resistors are combined to give the smallest possible resistance.

Parallel

[1]

- (ii) Fig. 13.2 shows one possible combination of the three resistors connected between the points A and B.

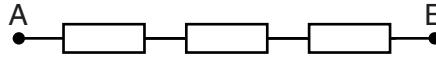


Fig. 13.2

In the three boxes of Fig. 13.3, draw three other possible combinations of all three resistors connected between points A and B. [2]

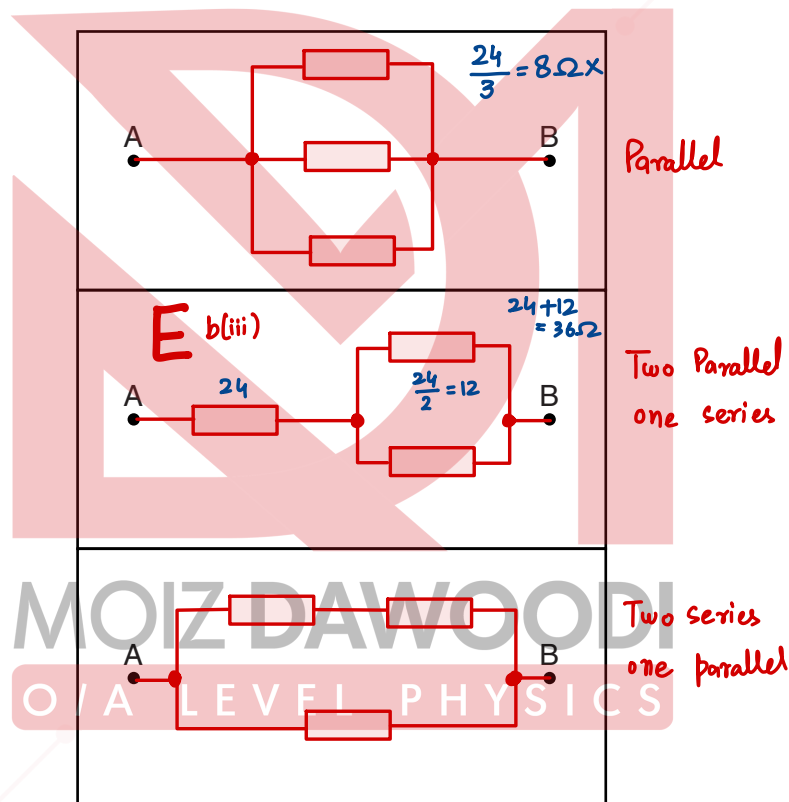


Fig. 3.3

- (iii) The student needs a resistance of 36Ω for his experiment. On Fig. 13.3, mark with the letter E the combination that has this resistance.

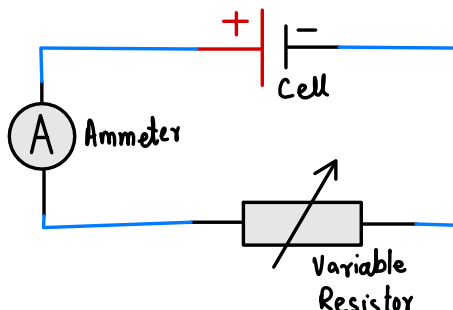
[1]

[ON2014/P42/Q3]

14 A student measures the time for which a 1.5 V cell can provide a steady current of 0.3 A.

(a) The student sets up a circuit including the cell. She measures the current produced by the cell and varies the resistance of the circuit to keep the current constant.

(i) In the space below, draw the circuit she uses.



[2]

(ii) Name the equipment the student uses to

1. measure the current,

Ammeter

2. vary the resistance,

Ohm meter

3. measure the time.

Stop watch

[1]

(iii) The student chooses from three analogue meters to measure the current. The meters have ranges 0 to 10 A, 0 to 1 A and 0 to 0.1 A.

Explain why the 0 to 1 A meter is the most suitable.

0-1 A can measure 0.3 A as accurately as possible.

[1]

(iv) Describe how the student adjusts the circuit to keep the current constant.

When resistance of circuit increases due to heating up, variable resistance is decrease so current may not change.

[1]

(b) A second student repeats the experiment using a much larger current. Suggest one hazard this may cause.

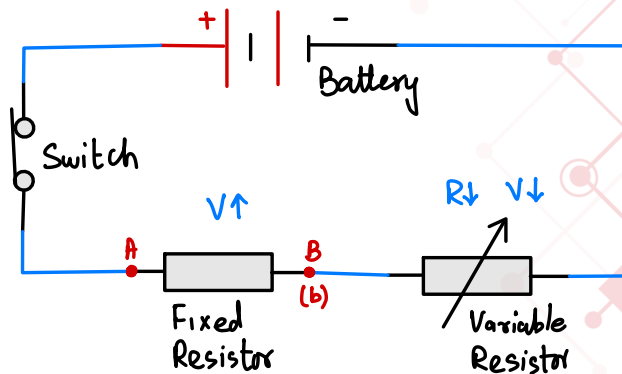
Overheating may cause circuit to catch fire.

[1]

[MJ2015/P41/Q4]

15 A student uses a cathode-ray oscilloscope (c.r.o.) to measure a voltage.

- (a) In the space below, draw a circuit diagram containing a battery, a switch, a fixed resistor and a variable resistor, all connected in series.



[1]

- (b) The c.r.o. is used to measure the voltage across the fixed resistor. On your circuit diagram, label the two points A and B where the c.r.o. is connected to the circuit.

[1]

- (c) Fig. 15.1a shows the c.r.o. screen when the switch in the circuit is open.

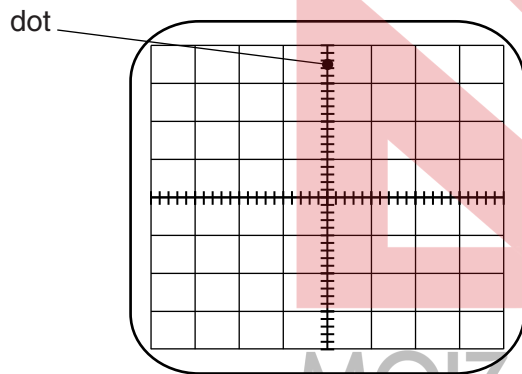


Fig. 15.1a

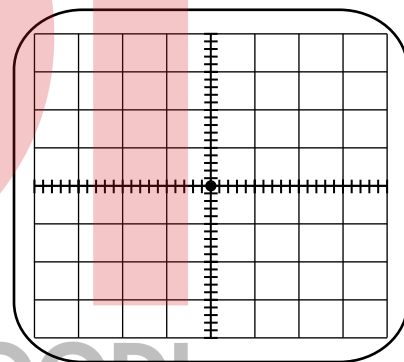


Fig. 15.1b

There are four control knobs on the c.r.o.:

X-shift
Y-shift
Y-gain
time-base

State which control knob is adjusted to move the dot to the centre of the screen, as shown in Fig. 15.1b.

Y-shift

[1]

- (d) The Y-gain is set at 2 V/div . With the dot in the centre of the screen, the switch is then closed and the dot moves to the position shown in Fig. 15.2.

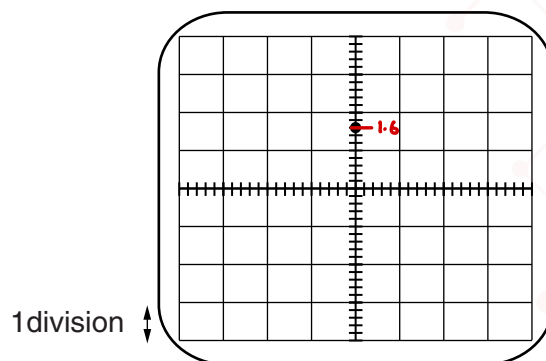


Fig. 15.2

- (i) Calculate the voltage V_{AB} across the fixed resistor.

$$\begin{aligned} 1 \text{ div} &\rightarrow 2\text{ V} \\ 1.6 \text{ div} &\rightarrow V \end{aligned}$$

$$V = \frac{2 \times 1.6}{1}$$

$$V_{AB} = 3.2\text{ V} \dots\dots\dots [1]$$

- (ii) The resistance of the variable resistor is reduced. State what happens to the dot on the screen.

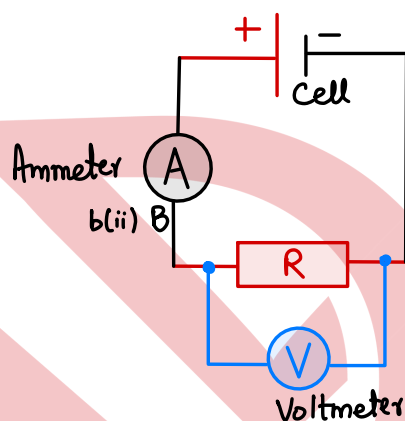
Trace will move upwards (as more voltage appears b/w AB).
 [1]

16 A student is asked to determine the resistance R of a resistor.

The student is provided with the following apparatus.

- the resistor with unknown resistance R
- four 1.5 V cells
- an ammeter
- a voltmeter
- connecting leads

(a) The student uses one of the 1.5 V cells in a circuit to determine the value of the resistance R . In the space below, draw the circuit diagram.



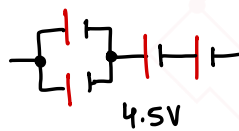
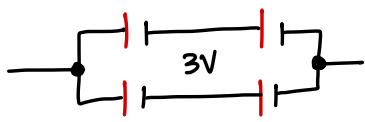
[2]

(b) The ammeter has one red terminal and one black terminal. The red terminal is marked '+'.
MOIZ DAWOODI
O/A LEVEL PHYSICS

(i) Explain why the terminals of the ammeter are different colours.

So that we know red terminal must be connected with positive & black terminal must be connected with negative side of cell. [1]

(ii) On your circuit diagram in (a), label, with the letter B, the black terminal of the ammeter. [1]



(c) The student has four 1.5 V cells.

(i) In addition to 1.5 V, state three other voltages that the student could use in her circuit.

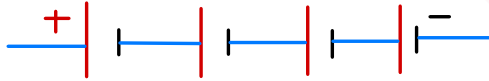
3.0 V

4.5 V

6.0 V

[1]

(ii) Draw the arrangement of cells that produces the largest voltage.



[1]

(d) The student repeats her experiment using the largest voltage.

State and explain whether this produces a different value for the resistance R .

Circuit will heat up & hence resistance increases.

[1]

[ON2015/P42/Q4]

MOIZ DAWOODI
O/A LEVEL PHYSICS

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- 3 (a) parallel B1 [1]
- (b) (i) correct voltmeter symbol drawn across power supply B1 [1]
(ii) X marked in series with resistor A B1 [1]
- (c) (i) 1.5V cao B1 [1]
(ii) 0.1(0) A ecf (c)(i) ÷ 15 B1 [1]
- (d) circuit 2 **PLUS**
two series resistors in parallel loop / no resistor in series with power supply
owtte / resistance is $6\frac{2}{3}\Omega$ B1 [1]

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- 3 (a) Y-shift **AND** X-shift B1 [1]
- (b) connections from either side of cell/resistor to Y inputs B1 [1]
- (c) (i) 2 V cao B1 [1]
(ii) 1.4 to 1.6 V B1 [1]
- (d) (i) dot moves down NOT to centre B1 [1]
(ii) dot moves horizontally/(dot becomes) (horizontal) line (on screen) B1 [1]

Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
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- 2 (a) (i) circuit with power supply and given wire with ammeter in series B1
variable resistor / variable power supply B1 [2]
- (ii) decrease variable resistor/resistance (of variable resistor) / increase supply voltage / increase number of cells B1 [1]
- (iii) reverse connections to battery/cell / change polarity of battery (accept reverse wire in the field) B1 [1]
- (iv) turn magnet other way up / S-pole on top and N-pole under wire / change polarity of magnets B1 [1]
- (b) wire becomes hot / melts / fuses / burns / trips power supply / damages/fuses ammeter B1 [1]

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3 (a) (i) A and B in series with cell and switch

B1

C labelled and in parallel with cell

B1 [2]

(ii) voltmeter across A

B1 [1]

(b)

Both lamp X and lamp Y are faulty	
Only lamp X is faulty	
Only lamp Y is faulty	
The cell is running down	✓
A connecting lead from the cell is broken	
The current in lamp X and lamp Y is too small	✓

B2 [2]

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4 (a) (i) S/south

B1 [1]

(ii) same as (a)(i)

B1 [1]

(iii) bar not magnetised/soft iron/
compass induces magnetism in bar/
N pole in centre of bar/
no keepers on magnet in drawer

B1 [1]

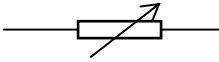
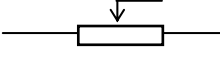
(b) marks may be awarded for clear diagram

in text	on diagram	
compass near magnet and mark end of plotting compass	magnet drawn with compass(near) with dot shown	M1
point to first mark, mark other end (along one field line)	new compass position along same correct field line OR series of dots along one correct field line	A1
continue to other pole or edge of paper OR join dots to give line OR repeat (to produce more field lines)	line of compasses or dots to other pole or edge of paper OR dots joined to give line OR more than 1 correct field line drawn	A1

[3]



Page 2	Mark Scheme	Syllabus	Paper
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- 2 (a) (i) lamp lights (normal brightness) B1 [1]
- (ii) any one from:
broken wire / connections not good
lamp blown / faulty
cell(s) run down B1 [1]
- (b) (i) lamp becomes dimmer B1 [1]
- (ii) 1. rheostat / variable resistor / potentiometer B1 [1]
2. correct circuit symbol drawn B1 [1]
-  
3. wire is coiled B1 [1]

Page 3	Mark Scheme	Syllabus	Paper
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- 2 (a) (i) 1. three resistors drawn in series B1 [1]
2. 470Ω cao unit required B1 [1]
- (ii) three resistors drawn in parallel B1 [1]

Page 4	Mark Scheme	Syllabus	Paper
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- 4 (a) (i) circuit diagram containing only solar cell, voltmeter and switch in series B1 [1]
- (ii) voltmeter terminals to wrong terminals of cell
current in voltmeter in wrong direction
voltmeter has polarity B1
- reverse connections to voltmeter
reverse connections to cell
connect red/+ve terminal of voltmeter to red/+ve terminal of cell B1 [2]
- (iii) needle drawn from centre to 0.96 V B1 [1]
- (b) (movement of) head/body reduces amount of light falling on solar cell
head/body not between window (light source) and cell B1
- sensible suggestion e.g.
position of solar cell/other light sources considered B1 [2]

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- 4 (a) circuit containing power supply, resistor, ammeter in series
voltmeter across resistor
means of varying current, e.g.
variable resistor
change additional resistor
variable power supply
change/add cells
use of potentiometer
B1
B1
B1
B1 [4]
- use of $V = IR$ and average
 I - V graph and find R
- (b) can be credited in (a) if seen
sensible suggestion, e.g.
check zero on meters
contacts tight / clean wires
avoid parallax in reading meters
keep current low
switch off between readings
use more sensitive meter
B1 [1]

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- 3 (a) (i) 0.9 V (unit required) B1 [1]
- (ii) crocodile clips
tight connections explained, e.g. wrap wire and tape B1 [1]
- (iii) same value/0.9 V and needle to right B1 [1]
- (b) sensible suggestion, e.g.
e.m.f./voltage too small
run down quickly/small amount of energy
voltage not steady
current too small
resistance too large B1 [1]
- (c) (i) 1. 2.7 (V) ecf 3 × (a)(i) B1 [1]
2. correct wiring in series **and** connected to voltmeter B1 [1]
- (ii) 1. 0.9 (V) ecf = (a)(i) B1 [1]
2. correct wiring in parallel **and** connected to voltmeter B1 [1]

Page 3	Mark Scheme	Syllabus	Paper
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- 2 (a) current cao [B1]
- (b) **any one from** [B1]
 (low resistance) does not decrease current (much)
 high resistance would decrease the current
 (low resistance) ammeter reads a large(r) value (than high R ammeter)
 current is high(er)
 very little p.d. across it
- (c) 0.67 A cao [B1]
- (d) **any one from** [B1]
 no parallax error
 needle does not stick
 easier to read / measure (current)
 easier to change range
 lower resistance
- (e) (i) **current** is same in series circuit/no junctions/single loop [B1]
- (ii) **any one from** [B1]
 meters not identical/exactly the same
 zero error in meter
 different calibration/calibration error

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- (b) (i) parallel B1 [1]
- (ii),(iii) any two correct combinations B1
 three correct combinations B1
 E marked correctly B1 [3]

Page 3	Mark Scheme	Syllabus	Paper
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- 4 (a) (i) correct circuit symbols for single cell, ammeter, variable resistor [B1]
 all three in series [B1]
- (ii) ammeter [B1]
 variable resistor/rheostat/potentiometer
 stopwatch/stop-clock/clock
 ALL THREE correct
- (iii) off scale of 0.1 A meter **and** [B1]
 10 A scale deflection too small
- (iv) reduce resistance (of variable resistor) (as current decreases) [B1]
- (b) cell/rheostat/wire becomes hot [B1]

Page 3	Mark Scheme	Syllabus	Paper
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- 3 (a) correct circuit symbols
all components in a series circuit B1
- (b) A and B labelled at ends of fixed resistor B1
- (c) y-shift c.a.o. no additions B1
- (d) (i) $3.2\text{V} \pm 0.2$ unit required B1
- (ii) dot moves up B1

Page 3	Mark Scheme	Syllabus	Paper
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- 4 (a) circuit containing **one** cell and resistor, with ammeter in series B1
- voltmeter in parallel with resistor/cell B1
- (b) (i) it matters which way round it is **connected**
so you can **connect** it the right way round B1
- (ii) ammeter terminal connected to –ve of cell labelled B B1
- (c) (i) 3 (V), 4.5 (V) and 6 (V) B1
- (ii) four cells drawn in series B1
- (d) no change/same value **and**
resistance independent of voltage/depends only on the resistor
OR
resistance increases **and**
resistor becomes hot B1